

Problem 3.22

If $A(3) = 100$ and $i_n = 0.02n$, find $A(6)$.

Solution

By definition,

$$i_n = 0.02n.$$

Therefore,

$$i_4 = 0.08, \quad i_5 = 0.10, \quad i_6 = 0.12.$$

Now use the recursion

$$A(n) = (1 + i_n)A(n - 1).$$

Starting from $A(3) = 100$,

$$A(4) = (1 + 0.08)(100) = 108.$$

Then,

$$A(5) = (1 + 0.10)(108) = 118.8.$$

Finally,

$$A(6) = (1 + 0.12)(118.8) = 133.056.$$

Hence,

$$A(6) \approx 133.06.$$